

Consider n random observations $X_1, \dots, X_n$ from a population with pmf/pdf f_θ , $\theta \in \Theta$.

The problem: To obtain a random interval $[L(X_1, \dots, X_n), U(X_1, \dots, X_n)]$ $\ni$ it contains the unknown parameter θ with some probability. So that, if,
 $P[L(X_1, \dots, X_n) \leq \theta \leq U(X_1, \dots, X_n)] \geq 1 - \alpha \quad \forall \theta \in \Theta$, and ^{some} $0 < \alpha < 1$ (usually, a small value).

We say, the random interval $[L, U]$ is a $100(1-\alpha)\%$ confidence interval (C.I.) for θ (or, C.I. for θ with confidence coefficient $(1-\alpha)$).

The '=' in the relationship holds for absolutely continuous distributions, and ' $\geq$ ' for discrete distributions, where '=' is not possible $\forall \alpha \in (0, 1)$.

Instead of θ , we may also obtain C.I. for some $\gamma(\theta)$, $\theta \in \Theta$.

Instead of a 2-sided C.I. $[L(X), U(X)]$ for $\gamma(\theta)$ (or, θ), we may also obtain one-sided C.I.s, like,

(i) $P[L(X) \leq \gamma(\theta)] \geq 1 - \alpha \quad \forall \theta \in \Theta$, then $[L(X), \infty)$ is a one-sided lower C.I.
 $100(1-\alpha)\%$ C.I. for $\gamma(\theta)$

(ii) $P[\gamma(\theta) \leq U(X)] \geq 1 - \alpha \quad \forall \theta \in \Theta$, then $(-\infty, U(X)]$ is a one-sided upper C.I.
 $100(1-\alpha)\%$ C.I. for $\gamma(\theta)$

Construction of C.I.s — using a pivot function

Suppose $\exists$ a statistic $T(X)$ and a function $\phi(T(X), \theta) \ni$ its distribution does not depend on θ , and

$$P[k_1 \leq \phi(T, \theta) \leq k_2] \geq 1 - \alpha \quad \forall \theta \in \Theta$$

Suppose, further, the pivot can be inverted $\ni$

$k_1 \leq \phi(T, \theta) \leq k_2 \Leftrightarrow c_1(T) \leq \gamma(\theta) \leq c_2(T)$ for some $\gamma(\theta)$ of θ & c_1, c_2 (fns of T only), we get the equivalent prob. statement, viz.,

$$P[c_1(T) \leq \gamma(\theta) \leq c_2(T)] \geq 1 - \alpha$$

$$P[C_1(T) \leq Y(\theta) \leq C_2(T)] \geq 1-\alpha \quad \forall \theta \in \Theta$$

Hence $[C_1(T(X)), C_2(T(X))]$ is a $100(1-\alpha)\%$ C.I. for $Y(\theta)$.

and further $[C_1^*(T(X)), C_2^*(T(X))]$ is a $100(1-\alpha)\%$ C.I. for θ .

Example 1. $X_1, \dots, X_n$ iid observations from $N(\mu, \sigma^2)$,
To get $100(1-\alpha)\%$ C.I. for μ .

Case (a) σ^2 known.

Note, $\bar{X} \sim N(\mu, \sigma^2/n)$, so that $Y = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \sim N(0,1)$ can work as a pivot.

$Y(X) = \phi(T(X, \sigma), \mu)$, $T(X, \sigma)$ is still a statistic, $\because \sigma$ is known.

So, $P(k_1 \leq Y \leq k_2) = \Phi(k_2) - \Phi(k_1) = 1-\alpha$ [Introducing $\Phi(\cdot)$ is possible because of introducing a pivot, which has $N(0,1)$ distn.]

$$\Rightarrow P\left[k_1 \leq \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \leq k_2\right] = P\left[\bar{X} - k_2 \frac{\sigma}{\sqrt{n}} \leq \mu \leq \bar{X} - k_1 \frac{\sigma}{\sqrt{n}}\right] = 1-\alpha$$

$$\Rightarrow P[C_1(\bar{X}) \leq \mu \leq C_2(\bar{X})] = 1-\alpha$$

This holds for infinitely many choices of (k_1, k_2) .

If possible, choose the one, which minimizes the length (or, expected length) under the given constraint.

In this case, length of the interval $= \frac{\sigma}{\sqrt{n}}(k_2 - k_1)$ & the constraint $\Phi(k_2) - \Phi(k_1) = 1-\alpha$.

Introduce the Lagrangian parameter λ to consider the following constrained minimization problem (wrt k_1, k_2)

$$L(k_1, k_2, \lambda) = \text{const.} \cdot (k_2 - k_1) + \lambda (\Phi(k_2) - \Phi(k_1) - (1-\alpha))$$

$$\text{Then } \begin{cases} \frac{\partial L}{\partial k_1} = -c - \lambda \phi(k_1) = 0 \Rightarrow \lambda = -\frac{c}{\phi(k_1)} \\ \frac{\partial L}{\partial k_2} = c + \lambda \phi(k_2) = 0 \Rightarrow \lambda = -\frac{c}{\phi(k_2)} \end{cases} \Rightarrow \phi(k_1) = \phi(k_2)$$

which holds for $k_2 = -k_1$ (in the present case)

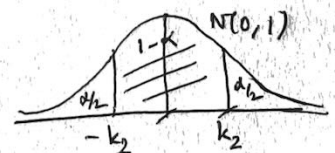
Thus, the shortest length C.I. is $\Rightarrow$

$$P\left[\bar{X} - k_2 \frac{\sigma}{\sqrt{n}} \leq \mu \leq \bar{X} + k_2 \frac{\sigma}{\sqrt{n}}\right] = 1-\alpha$$

$$\text{or, } P\left[\bar{X} - \frac{\sigma}{\sqrt{n}} z_{\alpha/2} \leq \mu \leq \bar{X} + \frac{\sigma}{\sqrt{n}} z_{\alpha/2}\right] = 1-\alpha$$

Hence, the shortest length $100(1-\alpha)\%$ C.I. for μ is

$$\bar{X} \pm \frac{\sigma}{\sqrt{n}} z_{\alpha/2}$$



But, for this case, we have a standard notation, viz. $z_{\alpha/2}$ for k_2 .